TreeSync Manual

TreeSync syncs two directory trees and checks every copy by SHA-256. It is four programs, in the **TreeSyncApp** folder of your app list:

- **TreeSyncBuild**: builds and tests it, with bootstrap's requirements check.
- **TreeSyncInstall**: installs the GCC and / or Clang release build for root, or removes it.
- **TreeSyncRoot-GCC**, **TreeSyncRoot-Clang**: sync as root, any folder (administrators).
- **TreeSyncHome-GCC**, **TreeSyncHome-Clang**: sync as you, inside your home (anyone).

A command-line program, `treesync-cli`, does the actual work; the two sync GUIs start it and show what it does.

This manual takes you from a fresh Linux machine to a running sync:

Part	What it covers
1. Bootstrap	Get the code, install the compilers and libraries, get U++, first build
2. Build	`build.sh`, TreeSyncBuild, debug and release, tests, `verify.sh` and the pre-push hook, TreeSyncInstall
3. Run	TreeSyncRoot and TreeSyncHome step by step, the command line, remote trees, root, checking and attributes
4. Files and troubleshooting	Where everything is kept, and what to do when something goes wrong

> **Warning.** TreeSync is experimental testing software. It copies, overwrites and deletes files. Try it on copies first, keep backups, and use **Dry run** before a real run. It is provided "AS IS" under the Apache License 2.0 (sections 7 and 8).

In TreeSync, the **Help** menu opens these chapters as PDFs, and **F1** opens the one for the program you are in (TreeSyncRoot and TreeSyncHome: Run; TreeSyncBuild and TreeSyncInstall: Build).

Rebuilding the PDFs. The `.md` files here are the sources. After editing them run `./build-pdf.sh` in this folder: it writes `pdf/`, one PDF per chapter and `TreeSync-Manual.pdf` with all of them (it needs python3 with its markdown module, Chrome or Chromium, and pdfunite).

Quick start

For someone who has done this before. Each step is explained in the parts above.

```
# 1. Get the code (dev-work, the default branch)
git clone https://github.com/JayACarlsonAtHome/jac317.git
cd jac317/v001

# 2. Check the machine, install what is missing (it asks first), build and test
./bootstrap.sh

# 3. Build the release programs (with all tests, both compilers)
./verify.sh

# 4. Install them for root (asks for your sudo password)
./install.sh

# 5. Start TreeSync: "TreeSync" in the desktop's app list, or
treesync
```

The installed TreeSync syncs as root for administrators (members of wheel); you build as yourself in `v001/`, and run `./install.sh` again after each build you want to use.

In the GUI: pick a **source** and a **target** folder, check what to sync in the source tree, then **Sync > Dry run** (Shift+F9) to see what would happen and **Sync > Run** (F9) to do it.

Words used in this manual

Word	Meaning
Source	The tree copied from
Target	The tree copied to
Copy / Mirror / Merge	The three modes: add and update only; make the target match the source (removing extras); copy changes both ways
Dry run	A run that only reports what it would do and changes nothing
Safe delete	Instead of deleting or overwriting, TreeSync moves the old file into `<target>.old`, or into `<target>/.treesync-old` when the target is the top folder of a disk (on by default)
Run	One sync, with a number; it is recorded in the database and can be resumed if stopped
Scan only	Reads and hashes both trees and compares them; changes nothing
`v001/`	The folder that holds the code, the scripts and (after a build) `out/` with the programs